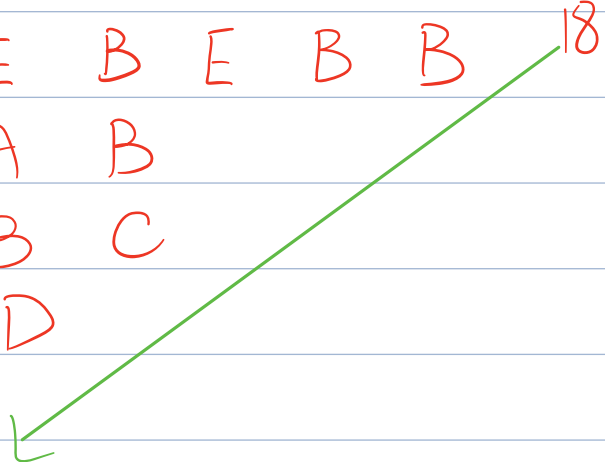


Ch 11

P1 E B E B B 18 D E
P2 A B
P3 B C
P4 D



18. (5 points)

- (a) (2 point) Melting point of *o*-nitrophenol is 44 °C and *p*-nitrophenol is 144 °C.
(b) (3 points) For *p*-nitrophenol hydrogen bonding is formed between two molecules, while for *o*-nitrophenol hydrogen bonding is formed inside one molecule. Intermolecular hydrogen bonding will increase the molecules' attraction, and intramolecular hydrogen bonding will decrease the molecules' attraction, so the melting point of *p*-nitrophenol is higher.

ch 13

P1 D D C
P2 D C D C C C
P3 B B C
P4 C D

$\text{LiBr} < \text{glucose} < \text{Zn}(\text{NO}_3)_2$

Ch 14

P1 A A B
P2 C A E E
P3 D C B
P4 A

16. (5 points)

- (a) (1 point) 3
(b) (2 points) 2. B and C are intermediates.
(c) (1 point) B → C step is rate determining.
(d) (1 point) For the overall reaction, ΔE is positive.

A D

20. (8 points)

(a)

$$k = Ae^{-E_a/RT} = 2.10 \times 10^{11} \times e^{-86.8 \times 1000 / (8.314 \times 313)} = 6.86 \times 10^{-4} M^{-1}s^{-1} \quad (2 \text{ Pts})$$

(b)

$$[KOH] = 0.343 \div (56.1 \times 0.279 \times 2) = 0.01096 M$$

$$[C_2H_5I] = 1.496 \div (156.0 \times 0.279 \times 2) = 0.01718 M$$

$$\text{Rate} = k[KOH][C_2H_5I] = 6.86 \times 10^{-4} \times 0.01096 \times 0.01718 = 1.29 \times 10^{-7} M/s \quad (4 \text{ Pts})$$

(c)

$$k = Ae^{-E_a/RT} = 2.10 \times 10^{11} \times e^{-86.8 \times 1000 / (8.314 \times 323)} = 1.93 \times 10^{-3} M^{-1}s^{-1} \quad (2 \text{ Pts})$$

OR

$$\ln k_2 = \ln k_1 - \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$= \ln (6.86 \times 10^{-4}) - \frac{86.8 \times 10^3}{8.314} \left(\frac{1}{323} - \frac{1}{313} \right) = 1.033$$

$$k_2 = 1.93 \times 10^{-3} M^{-1}s^{-1}$$

Ch 15

P1 C C D

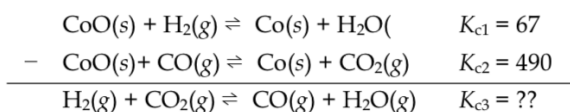
16: D

Part B short question (14 points)

17. (6 points)

- (a) right
- (b) left
- (c) left
- (d) no change
- (e) no change
- (f) right

18. (2 points)



$$K_{c3} = \frac{K_{c1}}{K_{c2}} = \frac{67}{490} = 0.137 = 0.14$$

18. (4 points)

(a) (1 pt)

$$K_p = \frac{P_{PCl_5(g)}}{P_{PCl_3(g)} P_{Cl_2(g)}} = \frac{1.30}{0.124 \times 0.157} = 66.8$$

(b) (1 pt) $K_p > 1$, thus the equilibrium favors products

(c) (2 pts)

$$K_p = K_c(RT)^{\Delta n} = 66.8$$
$$K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{66.8}{(0.08206 \times 450)^{-1}} = 2.50 \times 10^3$$

B

P2 B

Ch 16

P1 C D C E

P2 B B C

P3 D E D D

B3 (4 Pts)

(a) BrO^- (b) BrO^- (c) HPO_4^{2-} (d) F^-

B4 (4 Pts)

(b) ΔH is positive. Decreasing the temperature will shift the equilibrium in the direction that produces heat. In this case, the equilibrium shifts to the left, so the autoionization reaction of water is endothermic.

18. (4 points)

The value of pK_{a2} can only be choice (iii).

Calculate K_{a1} , assuming that only the first ionization determines pH.

$$[\text{H}_2\text{A}] = 0.10 \text{ M}; [\text{H}^+] = [\text{HA}^-] = 10^{-\text{pH}} = 10^{-3.30} = 5.0119 \times 10^{-4} = 5.0 \times 10^{-4}$$

$$K_{a1} = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} = \frac{(5.0119 \times 10^{-4})^2}{(0.10 - 5.0119 \times 10^{-4})} = \frac{(5.0119 \times 10^{-4})^2}{0.0995} = 2.524 \times 10^{-6} = 2.5 \times 10^{-6}$$

$\text{pK}_{a1} = 5.60$. pK_{a2} must be greater than pK_{a1} , which eliminates choices (i) and (ii).

If a solution of the salt NaHA is acidic, then $K_{a2} > K_{b1}$ and $\text{pK}_{a2} < \text{pK}_{b1}$.

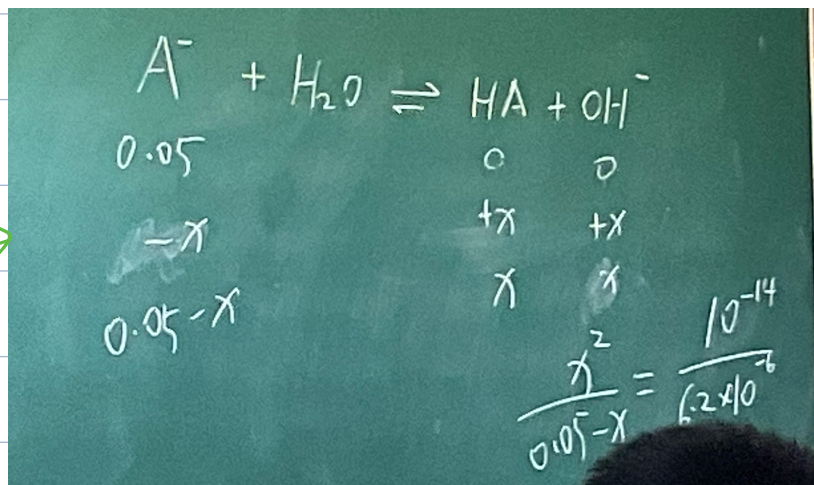
$$\text{pK}_{b1} = 14.00 - \text{pK}_{a1} = 14.00 - 5.60 = 8.40. \text{ This eliminates choice (iv).}$$

Ch 17

P1 B A

P2 E D D A E

P3 D



18. (6 points)

(a)

$$K_{sp} = [Ag^+][I^-] = 8.3 \times 10^{-17}$$

$$[Ag^+] = [I^-] = 9.11 \times 10^{-9} = 9.1 \times 10^{-9} M$$

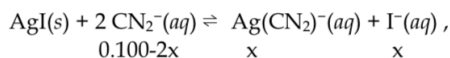
(b) calculate the equilibrium constant for the reaction $AgI(s) + 2 CN^-(aq) \rightleftharpoons Ag(CN)_2^-(aq) + I^-(aq)$,

$$K = K_{sp} \times K_f = 8.3 \times 10^{-17} \times 1.0 \times 10^{21} = 8.3 \times 10^4$$

(c)

K is much greater than one for the reaction of $AgI(s)$ with CN^- . This means that the reaction goes to completion. For a $AgI(s)$ in $0.100 M NaCN$ solution, CN^- is the limiting reactant. Two moles of CN^- react with one mole of AgI , so the solubility of AgI in $0.100 M NaCN$ is $(0.100/2) = 0.0500 M$.

OR



$$K = \frac{[Ag(CN_2)^-][I^-]}{[CN_2^-]^2} = \frac{x^2}{(0.100 - 2x)^2} = 8.3 \times 10^4$$

$$[I^-] = x = 0.05 M$$

D

21. (4 points)

$$\text{moles of HCl} = \frac{pV}{RT} = \frac{0.950 \text{ atm} \times 1.25 \text{ L}}{0.082 \text{ atm} \cdot \text{L/mol} \cdot \text{K} \times (273 + 25) \text{ K}} = 0.0486 \text{ mol}$$

$$\text{moles of NH}_3 = MV = 0.500 \text{ L} \times 0.150 \text{ M} = 0.075 \text{ mol}$$

	HCl(aq)	+	NH ₃ (aq)	→	NH ₄ Cl(aq)
Before reaction	0.0486 mol		0.075 mol		0
Change	-x		-x		x
After reaction	0		0.075-x		x
	x=0.0486 mol				

After reaction, there are $0.0264 \text{ mol NH}_3(aq)$ and $0.0486 \text{ mol NH}_4^+$ in the solution.

	NH ₃ (aq)	+	H ₂ O(l)	→	NH ₄ ⁺ (aq)	+	OH ⁻ (aq)
Initial concentration	0.0528 M				0.0972 M		0
Change concentration	-x				x		x
Eq. concentration	0.0528-x				0.0972+x		x
	$K_b(\text{NH}_3) = x(0.0972+x) / (0.0528-x) = 1.8 \times 10^{-5}$						
	x = $9.77 \times 10^{-6} M$						
	pOH = $-\log[\text{OH}^-] = 5.001$						
	pH = $14 - \text{pOH} = 8.990$						

ch 19

P1 A A

P2 A C E

P3 D D B E C

期末专块

11. C 14.

Increase: b c d e f g

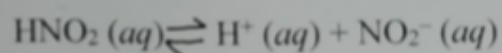
Decrease: a

A3. A

C2.

C2 (10 Pts)

(a) (2 Pts)



(b) (2 Pts)

1 / 2

2022 Spring

$$K_a = 4.5 \times 10^{-4}$$

$$\text{At equilibrium, } \Delta G^\circ = -RT \ln K_a = 8.314 \times 298 \times \ln(4.5 \times 10^{-4}) \div 1000 = \underline{19.1 \text{ kJ}}$$

(c) (3 Pts)

$$\text{At equilibrium, } \underline{\Delta G = 0}$$

(d) (3 Pts)

$$\Delta G = \Delta G^\circ + RT \ln Q$$

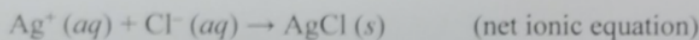
$$\Delta G = 19.1 + 8.314 \times 298 \times \ln(5.0 \times 10^{-2} \times 6.0 \times 10^{-4} \div 0.20) \div 1000 = \underline{-2.8 \text{ kJ} < 0}$$

So the forward reaction is spontaneous at 298K.

C3 (10 Pts)

C3 (10 Pts)

(a) (2 Pts)



(b) (2 Pts)

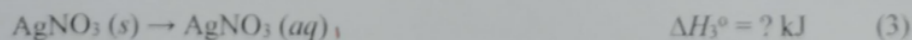
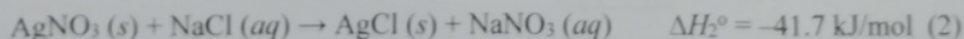
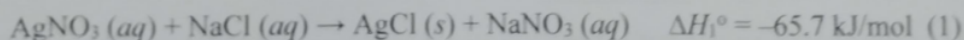
$$\begin{aligned} \Delta H^\circ &= \Delta H_f^\circ [\text{AgCl} (s)] - (\Delta H_f^\circ [\text{Ag}^+ (aq)] + \Delta H_f^\circ [\text{Cl}^- (aq)]) \\ &= -127.0 - (105.90 - 127.0) = \underline{-65.7 \text{ kJ/mol}} \end{aligned}$$

(c) (3 Pts)

Yes, I would expect ΔH° for the complete molecular equation will be the same as ΔH° for the net ionic equation.

Since the overall enthalpy change for the reaction is the enthalpy of products minus the enthalpy of reactants ($\Delta H_{\text{rxn}}^\circ = \sum n \Delta H_f^\circ (\text{products}) - \sum m \Delta H_f^\circ (\text{reactants})$), the contributions of spectator ions cancel.

(d) (3 Pts)



Using Hess's law, to get $\Delta H_3^\circ = \Delta H_2^\circ - \Delta H_1^\circ = 24 \text{ kJ/mol}$

For (3), $\Delta H_3^\circ = \Delta H_{\text{rxn}}^\circ = \Delta H_f^\circ [\text{AgNO}_3 (aq)] - \Delta H_f^\circ [\text{AgNO}_3 (s)]$

To get, $\Delta H_f^\circ [\text{AgNO}_3 (aq)] = \Delta H_3^\circ + \Delta H_f^\circ [\text{AgNO}_3 (s)] = 24 + (-124.4) = \underline{-100.4 \text{ kJ/mol}}$

So, ΔH_f° for $\text{AgNO}_3(aq)$ is -100.4 kJ/mol .

SAMPLE EXERCISE 19.10 Relating ΔG to a Phase Change at Equilibrium

(a) Write the chemical equation that defines the normal boiling point of liquid carbon tetrachloride, $\text{CCl}_4(l)$. (b) What is the value of ΔG° for the equilibrium in part (a)? (c) Use data from Appendix C and Equation 19.12 to estimate the normal boiling point of CCl_4 .

SOLUTION

Analyze (a) We must write a chemical equation that describes the physical equilibrium between liquid and gaseous CCl_4 at the normal boiling point. (b) We must determine the value of ΔG° for CCl_4 in equilibrium with its vapor at the normal boiling point. (c) We must estimate the normal boiling point of CCl_4 , based on available thermodynamic data.

Plan (a) The chemical equation is the change of state from liquid to gas. For (b), we need to analyze Equation 19.19 at equilibrium ($\Delta G = 0$), and for (c) we can use Equation 19.12 to calculate T when $\Delta G = 0$.

Solve

(a) The normal boiling point is the temperature at which a pure liquid is in equilibrium with its vapor at a pressure of 1 atm: $\text{CCl}_4(l) \rightleftharpoons \text{CCl}_4(g) \quad P = 1 \text{ atm}$

(b) At equilibrium, $\Delta G = 0$. In any normal boiling-point equilibrium, both liquid and vapor are in their standard state of pure liquid and vapor at 1 atm (Table 19.2). Consequently, $Q = 1$, $\ln Q = 0$, and $\Delta G = \Delta G^\circ$ for this process. We conclude that $\Delta G^\circ = 0$ for the equilibrium representing the normal boiling point of any liquid. (We would also find that $\Delta G^\circ = 0$ for the equilibria relevant to normal melting points and normal sublimation points.)

(c) Combining Equation 19.12 with the result from part (b), we see that the equality at the normal boiling point, T_b , of $\text{CCl}_4(l)$ (or any other pure liquid) is
Solving the equation for T_b , we obtain
Strictly speaking, we need the values of ΔH° and ΔS° for the $\text{CCl}_4(l)/\text{CCl}_4(g)$ equilibrium at the normal boiling point to do this calculation. However, we can *estimate* the boiling point by using the values of ΔH° and ΔS° for the phases of CCl_4 at 298 K, which we obtain from Appendix C and Equations 5.31 and 19.8:

$$\begin{aligned} \Delta G^\circ &= \Delta H^\circ - T_b \Delta S^\circ = 0 \\ T_b &= \Delta H^\circ / \Delta S^\circ \end{aligned}$$

$$\begin{aligned} \Delta H^\circ &= (1 \text{ mol})(-106.7 \text{ kJ/mol}) - (1 \text{ mol})(-139.3 \text{ kJ/mol}) = +32.6 \text{ kJ} \\ \Delta S^\circ &= (1 \text{ mol})(309.4 \text{ J/mol-K}) - (1 \text{ mol})(214.4 \text{ J/mol-K}) = +95.0 \text{ J/K} \end{aligned}$$

As expected, the process is endothermic ($\Delta H > 0$) and produces a gas, thus increasing the entropy ($\Delta S > 0$). We now use these values to estimate T_b for $\text{CCl}_4(l)$:

$$T_b = \frac{\Delta H^\circ}{\Delta S^\circ} = \left(\frac{32.6 \text{ kJ}}{95.0 \text{ J/K}} \right) \left(\frac{1000 \text{ J}}{1 \text{ kJ}} \right) = 343 \text{ K} = 70^\circ \text{C}$$

Note that we have used the conversion factor between joules and kilojoules to make the units of ΔH° and ΔS° match.

Check The experimental normal boiling point of $\text{CCl}_4(l)$ is 76.5°C . The small deviation of our estimate from the experimental value is due to the assumption that ΔH° and ΔS° do not change with temperature.